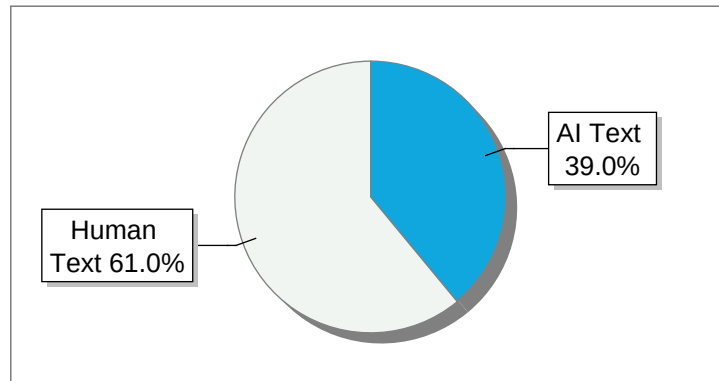


Submission Information

Author Name	Meghana
Title	SIGNAI: INDIAN SIGN LANGUAGE RECOGNITION USING VIDEOMAE-BASED VISION TRANSFORMERS
Paper--Submission ID	5670986
Submitted By	lib@pesce.ac.in
Submission Date	2026-05-18
Total [Pages,Sentences,Words]	[57, 684, 11901]
Document type	Article
Language	English

Result Information

AI Text: **39 %**



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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING P.

E. S.

College of Engineering, Mandya – 571 401 2025-26 A Project Report on “ SIGNAI: INDIAN SIGN LANGUAGE RECOGNITION USING VIDEO MAE- BASED VISION TRANSFORMERS” Under the guidance of Submitted in partial fulfillment for the award of Degree of Submitted by Dr.

RAKSHITH N Associate Professor, Dept.

of IS&E, PESCE Mandya.

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work “SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers” in partial

fulfillment for the award of Bachelor of Engineering of PES College of Engineering, Mandya (An Autonomous Institution, Affiliated to VTU, Belagavi) during the year 2025-26.

It is certified that all corrections indicated in internal assessment have been incorporated in the report deposited in the library.

The Project Work have been approved as it satisfies the academic requirements in respect of work prescribed for the degree in Bachelor of Engineering in Information Science and Engineering discipline.

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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING DECLARATION We, Chinmayee B

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P.E.S.

College of Engineering, Mandya, hereby declare that the work being presented in the project work entitled “SignAI: Indian Sign Language Recognition Using VideoMAE- Based Vision Transformers” is an authentic record of the work that has been independently carried out by us and submitted in partial fulfillment of the requirement for the award of degree in Bachelor of Engineering in Information Science and Engineering at P.E.S. College of Engineering during the academic year 2025-2026.

The work contained in this report has not been submitted to any other university or institution or professional body for the award of any other degree or any fellowship.

Date: Place: Mandya Chinmayee B [4PS22IS011] Dhanush patel M [4PS22IS018] Meghana G K [4PS22IS034]

Suman kumar matho [4PS23IS066] ACKNOWLEDGEMENT The satisfaction that accompanies the successful completion of the project report which would be complete only with the mention of the almighty god and the people who made it possible.

We are grateful to P.

E. S.

College of Engineering, Mandya for providing me an opportunity to enhance our knowledge through the project work.

We express our sincere thanks to Dr.

N.L. Murali Krishna, Principal, PESCE Mandya for providing us an opportunity and means to present the project work.

We express our heart full thanks to Dr.

T.S. Prabhakar, Associate Professor and Head, Department of Information Science and Engineering, PESCE

Mandya for encouragement in our project work, whose cooperation and guidance helped in nurturing this project report.

We would like to express profound thanks to the project guide Dr.

Rakshith N, Associate Professor, Department of Information Science and Engineering for the keen interest and encouragement in our project process.

Chinmayee B [4PS22IS011] Dhanush patel M [4PS22IS018] Meghana G K [4PS22IS034] Suman kumar matho [4PS23IS066] ABSTRACT Sign language is an essential mode of communication for the deaf and mute community, but it is not widely understood by the general population.

This creates a communication gap in daily interactions.

The aim of this project is to develop a system that can recognize Indian Sign Language (ISL) gestures from video sequences and convert them into meaningful text.

In this work, a deep learning-based approach is used to perform gesture recognition at the sentence level.

The system is built using a sequence-based model where each gesture is represented as a set of image frames.

The input video is processed into sequences of frames, which are resized and normalized before being passed to the model.

A Vision Transformer-based model, VideoMAE (base), is used for feature extraction and classification.

The model is fine-tuned using a pretrained version to adapt it for ISL gesture recognition.

The dataset consists of labeled gesture sequences, and each sequence is processed using a fixed number of frames.

The model learns both spatial and temporal features from the input data and predicts the corresponding gesture class.

The system is implemented using a full-stack architecture with a frontend for capturing input and a backend for processing and model inference.

The trained model achieves an accuracy of around 74% on the validation dataset.

The performance varies depending on gesture similarity and environmental conditions.

Although the system demonstrates the ability to recognize a wide range of gestures, certain limitations such as misclassification of similar gestures and challenges in real-time prediction are observed.

The project provides a foundation for further improvements in sign language recognition systems using advanced deep learning techniques.

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Communication is an important part of daily life, but for the deaf and mute community, it mainly depends on sign language.

Indian Sign Language (ISL) is widely used for communication, but it is not easily understood by people who are not trained in it.

This creates a communication gap between hearing-impaired individuals and the general population.

Sign language consists of both manual and non-manual features.

Manual features include hand gestures and movements, while non-manual features include facial expressions, eye movements, and head orientation.

Recognizing these gestures is a challenging task because it involves both spatial and temporal information.

With the advancement of computer vision and deep learning, it has become possible to develop systems that can automatically recognize sign language gestures from images and videos.

Earlier approaches mainly focused on using convolutional neural networks (CNN) and sequential models such as LSTM for gesture recognition.

However, these methods have limitations in capturing long-range dependencies in video sequences.

In this project, a deep learning-based system is developed to recognize Indian Sign Language gestures at the sentence level.

The system processes video input by converting it into sequences of frames.

Each sequence is then used as input to a transformer-based model known as VideoMAE.

This model is capable of learning both spatial and temporal features from the input data.

The dataset used in this project consists of labeled video sequences of ISL gestures.

Each sequence represents a sentence-level gesture, and the model is trained to classify these sequences into predefined gesture classes.

The system is implemented using a full-stack architecture, where the frontend handles input capture and the backend performs preprocessing and model inference.

The developed system demonstrates the ability to recognize a variety of ISL gestures with reasonable accuracy.

However, certain challenges such as similar gesture patterns and variations in real-world conditions affect the overall performance.

This project provides a foundation for further improvements in gesture recognition systems using advanced deep learning techniques.

CHAPTER-1

1.1 INTRODUCTION

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1.2 OBJECTIVES

The objectives of this project are defined to develop an effective Indian Sign Language recognition system using deep learning techniques.

The system focuses on processing video sequences and accurately identifying sentence-level gestures.

Objective 1 – To preprocess ISL video data into structured frame sequences for effective learning The first objective is to preprocess the sign language dataset and convert video data into structured frame sequences suitable for model input.

Each video is divided into ordered frames, resized, and normalized to ensure consistency.

Fixed-length sequences are formed to preserve temporal information in gestures, enabling the model to learn motion patterns effectively.

Objective 2 – To develop a VideoMAE-based model for recognizing sentence-level sign language gestures The second objective is to develop and train a deep learning model for gesture recognition using the VideoMAE architecture.

A pretrained VideoMAE model is fine-tuned on the ISL dataset to learn both spatial and temporal features from frame sequences.

The aim is to build a model capable of accurately classifying sentence-level gestures.

Objective 3 – To build an integrated system combining frontend interface and backend model inference The third objective is to design and implement a complete system that integrates frontend and backend components.

The frontend is responsible for capturing input through webcam or video upload, while the backend processes the frames and performs model inference.

The system generates the predicted gesture as text output for user interpretation.

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1.3 PROBLEM STATEMENT

Communication between the deaf and mute community and the general population is a major challenge due to the lack of understanding of sign language.

Indian Sign Language (ISL) is widely used for communication among hearing-impaired individuals, but most people are not familiar with it.

This creates a barrier in everyday interactions such as education, healthcare, and social communication.

Although several approaches have been developed for sign language recognition, many existing systems focus only on static gestures or limited vocabulary.

Recognizing sentence-level gestures from videos is more complex because it involves both spatial information (hand shapes and positions) and temporal information (movement across frames).

Variations in lighting, background, and gesture execution further increase the difficulty of accurate recognition.

Traditional models such as CNN and LSTM have been used for gesture recognition, but they often face limitations in capturing long-range dependencies in video sequences.

As a result, these methods may not perform well for complex and continuous gestures.

Therefore, there is a need to develop an efficient system that can process video sequences and accurately recognize Indian Sign Language gestures at the sentence level.

The system should be able to handle temporal variations in gestures and provide meaningful output in the form of text.

This project aims to address these challenges by developing a deep learning-based sign language recognition system using a transformer-based model.

The system focuses on improving gesture recognition performance by learning both spatial and temporal features from video data.

1.4 PROPOSED METHODOLOGY

The proposed methodology aims to develop an efficient system for recognizing Indian Sign Language (ISL) gestures from video input using a deep learning-based approach.

The system follows a structured pipeline that includes video processing, sequence formation, feature extraction

using a transformer-based model, and classification.

The process begins with video input provided by the user through the frontend interface.

The uploaded video is processed into individual frames, which are then encoded and transmitted to the backend server using an API request.

The backend, implemented using FastAPI, is responsible for handling frame decoding, preprocessing, and model inference.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 4 2025-26 Once the frames are received at the backend, they are decoded from base64 format and converted into RGB images.

These frames are then preprocessed to ensure uniformity across the dataset.

Each frame is resized to a fixed resolution of 224×224 pixels and normalized using ImageNet mean and standard deviation values.

This preprocessing step helps in maintaining consistency and improves the performance of the deep learning model.

After preprocessing, the frames are organized into sequences while preserving their temporal order.

A sliding window approach is used to select relevant frames from the video, ensuring that meaningful motion information is captured.

Each sequence is constructed with a fixed length of 16 frames, which serves as the input requirement for the model.

If a video contains more frames, it is trimmed accordingly, while maintaining the continuity of the gesture.

The prepared sequence is then passed to the VideoMAE model, which is a Vision Transformer-based architecture designed for video understanding tasks.

Unlike traditional models such as CNN and LSTM that process frames sequentially, VideoMAE treats the input as a spatiotemporal cube and applies self-attention mechanisms across all frames simultaneously.

This enables the model to learn both spatial features (such as hand shape and position) and temporal features (such as motion and transitions between frames).

In this project, a pretrained VideoMAE model is fine-tuned on the ISL dataset to improve its performance for gesture recognition.

Fine-tuning allows the model to adapt to domain-specific patterns while leveraging previously learned features.

The model outputs a set of class probabilities corresponding to different ISL sentence-level gestures.

The classification process is performed using a fully connected layer followed by a softmax function.

The final prediction is obtained by selecting the class with the highest probability using the argmax operation.

A confidence threshold is applied to filter out low-confidence predictions and improve reliability.

To ensure balanced learning, class weighting is used during training to handle class imbalance in the dataset.

The model is evaluated using performance metrics such as accuracy, precision, recall, and F1-score.

Additional evaluation methods such as confusion matrix and precision-recall analysis are used to gain deeper insights into model performance.

Overall, the proposed methodology provides a complete and effective pipeline for ISL gesture recognition by combining structured preprocessing, sequence modeling, transformer-based feature extraction, and classification.

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Figure 1.4.1 : Architecture of SIGNAL

Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 6 2025-26 Sign language recognition has gained significant attention in recent years due to its importance in bridging the communication gap between the deaf and mute community and the general population.

With the advancement of computer vision and deep learning techniques, researchers have explored various approaches to recognize gestures from images and videos.

Early methods mainly relied on handcrafted features and traditional machine learning models, but recent

developments focus on deep neural networks and transformer- based architectures.

Rastgoo et al.

[1] presented a comprehensive survey on sign language recognition using deep learning techniques.

The study highlighted the transition from traditional methods to deep learning-based approaches such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN).

The authors emphasized the importance of spatial and temporal feature learning for effective gesture recognition. However, the survey also pointed out challenges such as dataset limitations and difficulty in recognizing continuous gestures.

Koller et al.

[2] proposed a hybrid CNN-HMM model for continuous sign language recognition.

The system combines CNN for feature extraction and Hidden Markov Models (HMM) for temporal modeling.

This approach improved recognition accuracy for continuous gestures but required complex training and alignment of sequences, making it computationally expensive and less scalable.

Huang et al.

[3] introduced a video-based sign language recognition method that does not require temporal segmentation.

The model processes entire video sequences directly, reducing preprocessing complexity.

While the approach simplifies the pipeline, it may struggle with long sequences and complex gestures due to limited temporal modeling capability.

Camgoz et al.

[4] developed a transformer-based model for end-to-end sign language recognition and translation.

The use of transformers enabled the model to capture long-range dependencies in sequences, improving performance compared to traditional models.

However, the approach requires large-scale datasets and high computational resources for training.

Almasre and Al-Nuaim [5] proposed a real-time hand gesture recognition system using deep learning.

CHAPTER-2

2.1 LITERATURE REVIEW

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 7 2025-26 The model achieved good performance in controlled environments but faced challenges in real-world scenarios due to variations in lighting and background conditions.

Aly and Aly [6] developed a deep learning-based system capable of recognizing both static and dynamic gestures.

The model combined spatial and temporal features to improve accuracy.

However, the system required extensive training data and struggled with similar gesture patterns.

Ko et al.

[7] presented a CNN-based hand gesture recognition system focusing on image- based classification.

While the model achieved high accuracy for static gestures, it lacked the ability to capture temporal information, making it unsuitable for dynamic sign language recognition.

Kumar et al.

[8] focused on Indian Sign Language recognition using CNN models.

The study demonstrated the effectiveness of deep learning for ISL recognition but was limited to isolated gestures and did not address sentence-level recognition.

Pigou et al.

[9] explored CNN-based sign language recognition using image sequences.

The approach improved performance compared to single-frame methods but still faced limitations in capturing long-term dependencies across frames.

Shi et al.

[10] proposed a spatial-temporal network for continuous sign language recognition.

The model effectively captured motion patterns across frames, improving accuracy.

However, the complexity of the network increased computational requirements.

Zhou et al.

[11] introduced a multi-cue network that combines spatial and temporal features for gesture recognition. The model achieved better performance by integrating multiple feature representations but required careful tuning and large datasets.

Li et al.

[12] developed a deep neural network-based approach for vision-based sign language recognition. The study showed promising results but highlighted challenges in handling complex gestures and varying execution styles.

Li et al.

[13] proposed a framework for dynamic two-handed Indian Sign Language recognition. The model addressed challenges related to multi-hand gestures but required high-quality data and precise annotation for effective training.

Tamura et al.

[14] utilized YOLO-based object detection for hand gesture recognition. The model efficiently detected hand regions and improved recognition speed.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 8 2025-26 However, object detection alone is not sufficient for capturing temporal dependencies in gesture sequences.

Gupta and Sharma [15] developed a deep learning-based real-time sign language recognition system using computer vision techniques.

The system demonstrated practical usability but faced issues with accuracy in complex gestures and real-world conditions.

From the above literature, it is observed that earlier approaches mainly relied on CNN and sequential models such as LSTM or HMM, which have limitations in capturing long-range temporal dependencies.

Transformer-based models have shown significant improvements in handling sequence data, but they require efficient preprocessing and proper sequence modeling.

In this project, a transformer-based VideoMAE model is used to overcome these limitations.

Unlike traditional approaches, VideoMAE processes the entire sequence as a spatiotemporal representation and learns relationships across frames using self-attention.

This enables better recognition of dynamic gestures and sentence-level ISL data.

2.2 RESEARCH GAP

Based on the review of existing literature, several gaps can be identified in current sign language recognition systems.

Firstly, many existing systems focus on isolated gesture recognition rather than sentence-level understanding. Several approaches using CNN-based models are limited to recognizing static or single-word gestures and fail to capture continuous sentence-level expressions.

This restricts their applicability in real-world communication, where gestures are performed as sequences with temporal dependencies.

[7][8][9] Secondly, traditional models such as CNN combined with LSTM or HMM rely on sequential processing of frames, which limits their ability to capture long-range dependencies in video data.

These models process frames step-by-step and may lose important contextual relationships between distant frames, resulting in reduced accuracy for complex gestures.

[2] [10][11] Another major limitation is the lack of effective spatiotemporal feature learning.

While some models attempt to combine spatial and temporal features, they often require complex architectures and extensive tuning.

Additionally, these models struggle with variations in Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 9 2025-26 gesture execution, background conditions, and lighting, which affects their generalization ability in practical scenarios.

[5][6][12] Recent transformer-based approaches have shown improvements in sequence modeling, but they often require large-scale datasets and high computational resources.

Many existing works also do not focus specifically on Indian Sign Language (ISL), leading to a lack of robust

models for ISL sentence-level recognition.

[4][13] Furthermore, some approaches use object detection techniques such as YOLO for gesture recognition, which are effective for detecting hand regions but insufficient for capturing temporal dynamics of gestures. This limits their ability to recognize complex motion-based patterns in sign language.

[14] Another gap observed is the absence of a complete end-to-end system that integrates video processing, deep learning-based recognition, and user interface for practical usage.

Many studies focus only on model development without implementing a full system for real-world interaction.

[15] To address these limitations, the proposed project focuses on developing a sequence-based gesture recognition system using a transformer-based VideoMAE model.

The system processes video sequences as a whole, enabling effective learning of both spatial and temporal features.

Unlike traditional methods, the model uses self-attention mechanisms to capture relationships across all frames, improving recognition of sentence-level gestures.

The proposed system also provides a complete pipeline that includes video input, preprocessing, sequence formation, model inference, and output generation through a user interface.

By focusing on ISL sentence-level recognition and transformer-based learning, the project aims to overcome the limitations of existing approaches and improve overall gesture recognition performance.

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CHAPTER-3

3.1 SOFTWARE REQUIREMENTS

The development and deployment of the SignAI system require a combination of machine learning frameworks, programming tools, and web technologies to ensure efficient processing of video-based gesture recognition.

Operating System: The system is compatible with Windows or Linux-based environments (such as Ubuntu), providing flexibility for both development and deployment.

Programming Languages: Python is used for implementing the deep learning model and backend processing, while HTML, CSS, and JavaScript are used for frontend development.

Development Environment: Tools such as Visual Studio Code or Jupyter Notebook are used for model training, testing, and debugging.

Machine Learning Framework: PyTorch is used to implement and fine-tune the VideoMAE transformer model for gesture recognition tasks.

Computer Vision Library: OpenCV is used for video processing, frame extraction, resizing, and normalization.

Web Framework: FastAPI is used for backend API development to handle model inference and communication between frontend and backend.

Frontend Framework: React.js is used to build a user-friendly interface for uploading videos and displaying predictions.

Version Control System: Git is used for source code management and collaboration.

SYSTEM REQUIREMENTS AND ANALYSIS

3.2 HARDWARE REQUIREMENTS

The SignAI system requires appropriate hardware for efficient model training and inference.

Processor: Intel i5/i7 or equivalent processor for general development tasks.

GPU (Training): GPU is preferred for training the VideoMAE model, as it requires high computational power.

RAM: Minimum 8GB RAM is required, while 16GB or higher is recommended for smooth training and processing.

Storage: At least 10–20GB of storage is required to store dataset videos, trained models, and outputs.

Camera: A webcam can be used for capturing gesture videos (if extended to live input in future).

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3.3 SYSTEM ANALYSIS

The SignAI system is designed to bridge the communication gap between hearing-impaired individuals and normal users by automatically recognizing Indian Sign Language gestures from videos.

3.3.1 Existing System

Most existing systems rely on frame-based or image-based recognition, which fails to capture motion information.

Traditional models such as CNN or CNN-LSTM struggle with long-range temporal dependencies.

Many systems focus only on hand detection (YOLO-based) rather than full gesture understanding.

Limited datasets and variations in lighting and background reduce accuracy.

Real-time systems often lack robust sequence understanding.

3.3.2 Proposed System

The proposed SignAI system uses a transformer-based VideoMAE model to recognize ISL gestures from video sequences.

Converts videos into frame sequences (16 frames) Uses VideoMAE to learn spatial and temporal features

Applies self-attention mechanism for better sequence understanding Classifies gestures into 101 ISL classes

Provides text output (and optional voice output) This approach improves recognition of dynamic and complex gestures compared to traditional models.

3.3.3 Feasibility Analysis

Technical Feasibility: The system is feasible using existing deep learning frameworks like PyTorch and pretrained VideoMAE models.

Operational Feasibility: The system is user-friendly and can be easily used through a web interface.

Economic Feasibility: The system uses open-source tools and does not require expensive hardware except for training.

3.3.4 Advantages of Proposed System

Captures temporal motion using sequence-based learning Handles complex gestures effectively Uses transformer architecture for better accuracy Provides end-to-end pipeline (video → text output) Scalable for future improvements like real-time translation Dept of IS&E, PESCE, Mandya.

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3.4.1 Visual Studio Code

Visual Studio Code is used as the primary development environment for writing, debugging, and managing both frontend and backend code.

It supports multiple programming languages and extensions required for Python, React, and API development.

3.4.2 Python

Python is the core programming language used in this project.

It is utilized for: Implementing preprocessing steps such as frame extraction and normalization Training and fine-tuning the VideoMAE model Running inference on input videos Handling backend logic using FastAPI

3.4.3 FastAPI

FastAPI is used to build the backend server for the system.

It enables: Handling video upload requests Running model inference Sending prediction results back to the frontend Creating a scalable API-based architecture

3.4.4 PyTorch

PyTorch is used as the deep learning framework for implementing the VideoMAE model.

It provides flexibility for training transformer-based architectures and supports GPU acceleration for faster computation.

3.4.5 OpenCV

OpenCV is used for handling video and image processing tasks, including: Extracting frames from video sequences Resizing frames to 224×224 Normalizing pixel values Preparing data for model input

3.4.6 React.js

React.js is used to develop the frontend interface.

It provides: Video upload functionality Display of prediction results Visualization of model performance (accuracy, graphs) Interactive and user-friendly UI

3.4 DEVELOPMENT TOOLS

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3.5.1 Dataset Preparation

The system uses the ISL-CSLTR dataset, which contains: ~7,671 video sequences ~144,448 frames 101 gesture classes Each video is processed into structured frame sequences for model training.

3.5.2 Preprocessing and Augmentation

Before training, the dataset undergoes several preprocessing steps: Frame extraction from videos Resizing frames to 224×224 Normalization of pixel values Formation of fixed-length sequences (16 frames) Data augmentation techniques such as rotation, scaling, and flipping are applied to improve model generalization.

3.5.3 Model Training Framework

The VideoMAE model is trained using PyTorch with: Pretrained weights for transfer learning AdamW optimizer for efficient training Batch size of 8 and sequence length of 16 Evaluation metrics such as accuracy, precision, recall, and F1-score

3.5 DATA PROCESSING AND ANALYSIS

3.6.1 VideoMAE Transformer Model

The VideoMAE model is a transformer-based architecture designed for video understanding.

It: Divides frames into patches Uses self-attention to learn global dependencies Captures both spatial and temporal features Performs classification using a softmax layer

3.6.2 Model Inference Pipeline

During inference: User uploads a video through the frontend Frames are extracted and preprocessed The sequence is passed to the trained model

3.6 MACHINE LEARNING AND REAL-TIME INFERENCE

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 14 2025-26 The model predicts the gesture class Output is displayed as text and voice output

3.6.3 Performance Optimization

To improve performance: Pretrained VideoMAE model is fine-tuned Efficient sequence-based processing is used GPU acceleration is utilized during training Optimized API responses ensure faster inference Dept of IS&E, PESCE, Mandya.

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CHAPTER-4

4.1 INTRODUCTION

Sequence-based data processing plays a crucial role in video-based sign language recognition systems.

Unlike image-based approaches that rely on single-frame inputs, sign language involves continuous motion, where meaning is conveyed through a sequence of gestures over time.

Therefore, it is essential to process video data in a structured manner that preserves temporal information.

In this project, the focus is on converting raw video input into meaningful frame sequences that can be effectively used for deep learning models.

The input video is first divided into individual frames, which are then organized in a sequential order.

Maintaining this temporal order is important because the relationship between consecutive frames helps in understanding gesture dynamics.

To ensure consistency across the dataset, each frame is resized to a fixed resolution and normalized.

This preprocessing step reduces variations caused by differences in lighting, background, and camera quality.

After preprocessing, the frames are grouped into fixed-length sequences.

A sequence length of 16 frames is used, as it provides a balance between capturing sufficient motion information and maintaining computational efficiency.

A sliding window approach is applied to generate multiple sequences from a single video.

This technique helps in utilizing the data more effectively and ensures that different parts of the gesture are captured.

It also increases the number of training samples, which improves the performance of the model.

The processed sequences are then used as input to the deep learning model.

By representing gestures as sequences rather than individual frames, the system is able to capture both spatial and temporal features.

This approach forms the foundation for accurate gesture recognition in the proposed system.

Overall, sequence-based data processing is a key step in the system, as it directly influences the quality of input provided to the model and ultimately affects the performance of gesture recognition.

PREPROCESS ISL VIDEOS INTO FRAME SEQUENCES FOR MODEL TRAINING Dept of IS&E, PESCE, Mandya.

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4.2 DATASET DESCRIPTION

The dataset used in this project is based on Indian Sign Language (ISL) and is designed for sentence-level gesture recognition.

It consists of video sequences representing different ISL sentences performed by multiple signers.

The dataset provides a diverse set of gestures, which helps in training the model to recognize variations in gesture execution.

The dataset contains a total of 7,671 video sequences with approximately 144,448 frames distributed across 101 gesture classes.

Each class corresponds to a specific ISL sentence, such as common daily communication phrases.

The data is organized in a structured format where each class contains multiple sequences, and each sequence consists of ordered image frames.

To prepare the data for training, videos are converted into frame sequences.

A sliding window approach is used to generate fixed-length sequences from continuous frames.

Each sequence contains 16 frames, which are resized to a resolution of 224×224 pixels.

This ensures compatibility with the deep learning model and maintains consistency across all samples.

The dataset is further divided into training, validation, and testing sets to evaluate model performance effectively. The splitting is performed in a class-balanced manner to ensure that each gesture category is properly represented in all subsets. This helps the model generalize better and prevents bias toward specific classes.

This repository can be accessed at: ISL- CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition - Mendeley Data, accessed on 2 December 2024.

Overall, the dataset provides a strong foundation for training and evaluating the proposed system, as it captures both spatial and temporal aspects of sign language gestures.

The proposed methodology focuses on processing video data into structured sequences for effective sign language recognition.

Since sign language is inherently temporal, capturing motion across frames is essential for accurate interpretation.

The system begins with raw video input, which is converted into individual frames using frame extraction techniques.

These frames are arranged in temporal order to preserve the continuity of gestures.

Each frame is then resized to a fixed resolution of 224×224 pixels and normalized using ImageNet mean and standard deviation values to ensure consistency across the dataset.

A sliding window approach is applied to divide the continuous frames into smaller segments.

This enables efficient utilization of video data and allows the model to learn from different parts of the gesture.

4.3 PROPOSED METHODOLOGY

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From these segments, fixed-length sequences of 16 frames are formed.

These sequences capture both spatial and temporal features of gestures and serve as the input for the deep learning model.

The final output of this process is a structured sequence dataset that is ready for training, validation, and testing. This methodology ensures that meaningful motion information is preserved and provides a strong foundation for accurate gesture recognition.

Figure 4.3.1 : Sequence-Based Data Processing Architecture

1.

Raw Video Input The process begins with collecting raw video data representing Indian Sign Language gestures. Each video contains continuous motion where gestures are performed sequentially. These videos act as the primary input for the system.

2. **Frame Extraction** The input video is divided into individual frames using video processing techniques. Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 18 2025-26 Each frame represents a snapshot of the gesture at a particular time instance.

This step converts the video into image data, which can be further processed by the model.

3. **Frame Sorting (Temporal Order)** After extraction, frames are arranged in the correct temporal order. Maintaining the sequence of frames is important because gesture meaning depends on motion across time. Incorrect ordering may lead to loss of gesture information.

4. **Resize Frames (224×224)** Each frame is resized to a fixed resolution of 224×224 pixels.

This ensures uniform input size across the dataset and makes the data compatible with the VideoMAE model, which requires fixed input dimensions.

5. **Normalization (ImageNet Mean & Standard Deviation)** The resized frames are normalized using ImageNet mean and standard deviation values.

This step standardizes pixel values and improves model convergence during training.

It also helps the pretrained model adapt better to the new dataset.

6. Sliding Window Processing A sliding window technique is applied to generate multiple sequences from a single video.

Instead of using the entire video at once, smaller overlapping segments are extracted.

This improves data utilization and allows the model to learn different parts of the gesture effectively.

7. Sequence Formation (16 Frames) From the processed frames, fixed-length sequences of 16 frames are created.

This sequence length is chosen to balance temporal information and computational efficiency.

Each sequence represents a complete gesture segment.

8. Sequence Dataset (Ready for Model Training) The final sequences are stored in a structured format, where each class contains multiple sequences.

This dataset is then used for training, validation, and testing of the deep learning model.

Each sequence serves as an input sample for the model.

4.4 RESULT AND CONCLUSION

4.4.1 Experimental Results

The experimental results for Objective 1 focus on validating the effectiveness of the sequence-based data processing pipeline.

The primary goal of this stage is to convert raw Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 19 2025-26 video input into structured sequences suitable for model training while preserving temporal information.

The implemented pipeline successfully extracts frames from input videos and maintains their correct temporal order.

The preprocessing steps, including resizing to 224×224 resolution and normalization using ImageNet parameters, ensure uniformity across all frames.

This standardization helps in reducing variations caused by different recording conditions such as lighting and background.

The use of sliding window processing enables efficient utilization of video data by generating multiple overlapping sequences from a single video.

This increases the number of training samples and allows the model to learn different segments of gestures effectively.

The sequence formation process ensures that each input sample consists of a fixed length of 16 frames, which satisfies the requirements of the VideoMAE model.

The generated dataset is well-structured and balanced across gesture classes, making it suitable for training, validation, and testing.

The consistency in sequence length and preprocessing contributes to stable input representation, which is essential for improving model performance in later stages.

Overall, the results demonstrate that the data processing pipeline is capable of transforming raw video data into meaningful and model-ready sequences.

Figure 4.4.1.1 : Training log showing successful preprocessing, sequence

formation, and model training using ISL video frame sequences Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 20 2025-26 The sequence-based data processing pipeline effectively transforms raw video input into structured and model-ready sequences while preserving essential temporal information.

By performing frame extraction, resizing, normalization, and sequence formation, the system ensures

consistency across all input samples.

The use of a sliding window approach further enhances data utilization by generating multiple meaningful sequences from a single video, thereby improving the diversity of training data.

Overall, this methodology provides a reliable foundation for deep learning-based sign language recognition. The structured representation of gestures as fixed-length sequences makes the data suitable for transformer-based models such as VideoMAE.

This preprocessing pipeline plays a crucial role in improving model performance and ensures that both spatial and temporal features are accurately captured for effective gesture understanding.

4.4.2 Conclusion

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 21 2025-26 Model development is a critical stage in sign language recognition systems, as it determines how effectively spatial and temporal features are learned from input data.

Unlike traditional image-based approaches, video-based gesture recognition requires capturing both the appearance of hand gestures and their motion across time.

To address this, transformer-based architectures have gained attention due to their ability to model long-range dependencies in sequential data.

In this project, the VideoMAE (Video Masked Autoencoder) model is used for gesture recognition.

VideoMAE is a Vision Transformer-based model specifically designed for video understanding tasks.

It processes video sequences as a unified spatiotemporal representation rather than analyzing frames individually.

This allows the model to capture relationships between frames more effectively compared to conventional methods such as CNN or CNN-LSTM.

The model takes input in the form of fixed-length frame sequences, where each sequence consists of 16 frames of size 224×224 pixels.

These sequences are divided into smaller patches, and self-attention mechanisms are applied to learn meaningful features across both spatial and temporal dimensions.

By leveraging this approach, the model can identify subtle variations in gestures and motion patterns.

A pretrained VideoMAE model is fine-tuned on the ISL dataset to adapt it for sentence-level gesture classification.

Fine-tuning helps in improving performance by utilizing previously learned representations while adjusting the model to the specific characteristics of sign language data.

Overall, this objective focuses on developing an efficient and robust model capable of recognizing ISL gestures by combining transformer-based feature learning with structured sequence input.

CHAPTER-5

5.1 INTRODUCTION

DEVELOP A VIDEOMAE MODEL FOR SENTENCE- LEVEL GESTURE RECOGNITION Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 22 2025-26 The proposed methodology focuses on recognizing Indian Sign Language (ISL) gestures from video sequences using a transformer-based VideoMAE model.

Unlike traditional image-based approaches, this system processes gesture data as a sequence of frames, allowing it to capture both spatial and temporal information.

Sign language is inherently dynamic, where the meaning of a gesture depends not only on hand shape but also on motion over time.

Therefore, analyzing frames independently is not sufficient for accurate recognition.

In this approach, input video sequences are first processed and converted into structured frame sequences.

These sequences are then used as input to the VideoMAE model, which learns meaningful representations using self-attention mechanisms.

A pretrained VideoMAE model is fine-tuned on the ISL dataset to improve its performance for sentence-level gesture classification. The model outputs probability scores for each gesture class, and the final prediction is obtained using a softmax classification layer. This methodology ensures efficient learning of spatio-temporal features and provides a robust solution for recognizing complex ISL gestures.

5.3 PROPOSED METHODOLOGY

Figure 5.3.1 : VideoMAE-based Spatio-Temporal Feature Extraction and Classification Architecture
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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 23 2025-26 1.

Raw Video Input The process begins with raw video input representing ISL gestures.

Each video contains a sequence of frames capturing the motion of hand gestures over time.

Unlike static image recognition, video input provides temporal continuity, which is essential for understanding gestures correctly.

These videos are processed using a sliding window approach, where segments of frames are extracted for further processing.

This ensures that different parts of the gesture are captured and used for training.

2. Frame Representation and Temporal Structure After extraction, frames are organized in temporal order.

The sequence maintains the natural flow of the gesture, which is crucial for capturing motion patterns.

Each frame is resized to 224×224 pixels to maintain uniformity and compatibility with the model.

The temporal structure of frames allows the system to treat the input as a spatio-temporal volume rather than independent images.

This improves the model's ability to learn gesture dynamics.

3. Patch Embedding In the next stage, each frame is divided into smaller patches, typically of size 16×16 pixels.

These patches act as the fundamental units of input for the transformer model.

Each patch is flattened and converted into a vector representation called an embedding.

This process transforms visual information into numerical form, making it suitable for processing by the transformer architecture.

Patch embedding reduces computational complexity while preserving important spatial details.

4. Transformer Encoder (Self-Attention Mechanism) The embedded patches are passed through multiple transformer layers.

Each layer uses a self-attention mechanism to learn relationships between different patches across all frames.

This is a key advantage over traditional models.

Unlike CNNs, which focus on local features, or LSTMs, which process sequences sequentially, transformers can analyze all patches simultaneously.

This enables the model to capture long-range dependencies and interactions between different parts of the video.

Through this process, the model learns both: Spatial features (hand shape, position, orientation) Temporal features (movement, transitions between gestures) This combined understanding is essential for accurate sign language recognition.

5. Feature Representation After passing through the transformer encoder, the model generates a rich feature representation of the input sequence.

This representation encodes all relevant information about the gesture, including both appearance and motion.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 24 2025-26 6.

Decoder (Conceptual Component) The diagram shows a decoder component, which is part of the original VideoMAE architecture.

However, in this project, the focus is on using a pretrained VideoMAE model for classification rather than reconstruction.

Therefore, the decoder is not actively used for reconstructing masked patches but can be considered as part of the internal architecture.

The primary role of the system is feature extraction and classification.

7. Classification Head (Softmax Layer) The extracted features are passed to a classification head, which consists of a fully connected layer followed by a softmax function.

This layer converts the learned features into probability scores for each gesture class.

The softmax function ensures that the output values represent probabilities, making it easier to interpret the results.

8. Final Prediction (Argmax Selection) The final predicted class is obtained by selecting the class with the highest probability using the argmax function.

This class corresponds to the recognized ISL sentence or gesture.

This step converts the model's output into a meaningful label that can be understood by the user.

9. Sentence-Level Gesture Output The output of the system is a sentence-level gesture label.

Unlike systems that recognize only individual signs, this approach focuses on recognizing complete gestures representing meaningful phrases.

This improves usability and makes the system more suitable for real-world communication scenarios.

The proposed methodology combines sequence-based data processing with transformer-based feature extraction to achieve effective sign language recognition.

By using VideoMAE, the system overcomes limitations of traditional models and captures both spatial and temporal aspects of gestures.

The structured pipeline ensures that input data is processed efficiently, and the model is able to learn meaningful patterns for accurate classification.

This approach provides a strong foundation for building intelligent and scalable sign language recognition systems.

5.3 RESULT AND CONCLUSION

5.3.1 Experimental Results

A comparative analysis of the proposed VideoMAE-based model demonstrates its effectiveness in recognizing Indian Sign Language (ISL) gestures from video sequences.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 25 2025-26 The model is capable of learning both spatial and temporal features, which are essential for accurate gesture recognition.

However, challenges still exist due to variations in gesture execution, background conditions, and similarities between certain gesture classes.

The experimental results indicate that transformer-based models such as VideoMAE provide improved performance compared to traditional approaches like CNN, LSTM, and classical machine learning models. These earlier models often struggle to capture long-range temporal dependencies, whereas VideoMAE effectively models the entire sequence using self-attention mechanisms.

Despite this improvement, some misclassifications occur due to overlapping gesture patterns and limited dataset diversity.

Figures 5.3.1, 5.3.2, and 5.3.3 illustrate the performance of the proposed model on the ISL dataset.

Figure 5.3.1 presents the confusion matrix, which shows that most gesture classes are correctly classified, indicated by strong diagonal values.

Some off-diagonal values represent misclassifications between visually similar gestures, which is expected in complex multi-class problems.

Figure 5.3.2 shows the training and validation accuracy over multiple epochs. The graph indicates a steady improvement in accuracy during training, with the model achieving good convergence. The validation accuracy follows a similar trend, confirming that the model generalizes well to unseen data.

Figure 5.3.3 depicts the training and validation loss curves. The decreasing trend in loss values demonstrates effective learning.

However, a slight gap between training and validation loss suggests mild overfitting, which can be addressed through further regularization and data augmentation.

Additionally, precision-recall analysis (Figure 5.3.4) highlights the model's ability to balance precision and recall across multiple gesture classes.

The results confirm that the model performs consistently across different categories.

Overall, the experimental outcomes validate the effectiveness of transformer-based approaches for video-based sign language recognition while also highlighting challenges such as gesture similarity and dataset limitations.

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Figure 5.3.1 : Confusion Matrix of VideoMAE Model for ISL Gesture Classification

Figure 5.3.2 : Training and Validation Accuracy of VideoMAE Model over Epochs

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Figure 5.3.3 : Training and Validation Loss of VideoMAE Model over Epochs.

Figure 5.3.4 : Precision–Recall Curve for Multi-Class ISL Gesture Classification.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 28 2025-26 The results demonstrate that the proposed VideoMAE-based model is capable of effectively recognizing ISL gestures by leveraging spatio-temporal feature learning.

The use of transformer architecture allows the model to capture complex motion patterns, leading to improved performance compared to traditional methods.

The evaluation metrics and graphical analysis confirm that the model achieves reliable accuracy and generalization.

However, certain limitations such as misclassification of similar gestures and mild overfitting indicate scope for further improvement.

Overall, the proposed approach provides a strong foundation for developing advanced sign language recognition systems and can be enhanced with larger datasets and improved optimization techniques.

5.3.2 Conclusion

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 29 2025-26 The integration of deep learning with web-based systems has enabled the development of intelligent applications that can perform complex tasks such as gesture recognition and real-time communication.

In this project, a complete system has been designed and implemented to recognize Indian Sign Language (ISL) gestures using a video-based input pipeline.

The system combines a user-friendly frontend interface with a powerful backend model to deliver accurate and

efficient predictions.

The frontend of the system is developed using React, which provides an interactive and responsive user interface.

It allows users to upload video files containing sign language gestures.

Once a video is uploaded, the interface displays the video preview and initiates the prediction process.

The design ensures smooth user interaction and clearly presents intermediate stages such as processing status, prediction output, and confidence score.

The backend is implemented using FastAPI, which handles all processing and model inference tasks.

The uploaded video is sent to the backend through API requests, where it undergoes frame extraction, preprocessing, and sequence formation.

These processed frames are then passed to the VideoMAE deep learning model, which extracts spatial and temporal features to recognize the gesture.

The model outputs the predicted class along with a confidence score, which is returned to the frontend.

The system further enhances usability by converting the predicted gesture into readable text and optional speech output.

This enables better accessibility and supports communication for users who rely on sign language.

The integration of frontend and backend components ensures seamless data flow, efficient processing, and real-time user feedback.

Overall, this objective focuses on building a complete, end-to-end system that connects user interaction with advanced deep learning models.

CHAPTER-6

6.1 INTRODUCTION

BUILD A COMPLETE FRONTEND AND BACKEND SIGN LANGUAGE RECOGNITION SYSTEM Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 30 2025-26 The proposed system follows an end-to-end pipeline that integrates a frontend interface with a backend deep learning model to perform Indian Sign Language (ISL) gesture recognition from uploaded videos.

The methodology is designed to ensure smooth data flow, efficient preprocessing, and accurate prediction using sequence-based learning.

The process begins at the frontend, where the user uploads a video containing sign language gestures through a React-based interface.

The uploaded video is previewed on the screen, and upon user action, it is sent to the backend using an API request.

The video data is encoded and transmitted to the FastAPI server for further processing.

At the backend, the received video is first decoded and converted into individual frames.

These frames are then arranged in temporal order to preserve the motion information present in the gesture.

Each frame undergoes preprocessing, which includes resizing to a fixed resolution of 224×224 pixels and normalization using standard ImageNet mean and standard deviation values.

This ensures consistency in input format and improves model performance.

After preprocessing, the system performs sequence formation by selecting a fixed number of frames (16 frames) from the video.

This step is essential as the model operates on temporal sequences rather than individual images.

The sequence captures the dynamic nature of gestures, enabling the model to learn both spatial and temporal features effectively.

The processed sequence is then passed to the VideoMAE model, which is a transformer-based architecture designed for video understanding tasks.

The model extracts meaningful features by analyzing spatial patches and temporal dependencies across frames.

A classification head is applied on top of the extracted features, which produces probability scores for each gesture class using a softmax layer.

6.2 PROPOSED METHODOLOGY

The implementation demonstrates how modern web technologies and AI models can be combined to create practical and impactful assistive applications.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 31 2025-26 The class with the highest probability is selected as the final prediction using the argmax function.

Additionally, a confidence score is calculated to indicate the reliability of the prediction.

This predicted gesture is then converted into text and sent back to the frontend.

Finally, the frontend displays the predicted text output along with the confidence score.

An optional speech output feature is also provided to enhance user accessibility.

This complete methodology ensures accurate gesture recognition through effective integration of user interface, data processing, and deep learning-based classification.

Figure 6.2.1 : End-to-end workflow of the proposed SignAI system for Indian Sign Language recognition 1.

User Uploads Video The process begins when the user uploads a sign language video through the React- based frontend interface.

The uploaded video contains ISL gesture sequences to be recognized by the system.

2. Video Preview After uploading, the selected video is displayed on the frontend screen.

This allows the user to verify the uploaded gesture video before processing.

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Video Sent to Backend Once the user starts prediction, the frontend sends the video to the FastAPI backend through an API request for further processing and model inference.

4 .

Video Reception at Backend The FastAPI server receives the uploaded video and prepares it for preprocessing and gesture recognition.

5. Frame Extraction The uploaded video is converted into individual frames while maintaining the correct temporal order.

These frames represent the motion information present in the gesture.

6. Frame Preprocessing Each extracted frame is resized to 224×224 pixels and normalized using ImageNet statistics.

This preprocessing ensures consistency in input format and improves model performance.

7. Sequence Formation A fixed sequence of 16 frames is selected from the video.

The sequence-based representation helps the model learn temporal relationships and gesture continuity.

8. Input to VideoMAE Model The generated frame sequence is passed to the transformer-based VideoMAE model, which learns spatial and temporal features from the gesture sequence.

9. Classification The extracted features are processed through a Softmax classification layer, which generates probability scores for all gesture classes.

10. Final Prediction The class with the highest probability is selected using the argmax function.

A confidence score is also generated to indicate prediction reliability.

11. Text and Speech Output The predicted gesture is converted into text and sent back to the frontend.

The system can also generate optional speech output for improved accessibility.

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6.3 RESULT AND CONCLUSION

6.3.1 Experimental Results

The developed system was evaluated based on its ability to process uploaded videos and generate correct Indian Sign Language (ISL) gesture predictions.

The results demonstrate that the integration of a React-based frontend with a FastAPI backend and VideoMAE model provides a smooth and functional pipeline for gesture recognition.

The system successfully processes input videos, extracts frames, performs sequence-based inference, and generates meaningful outputs in the form of text and speech.

Figures 6.3.1, 6.3.2, and 6.3.3 illustrate the working and output of the implemented system.

Figure 6.3.1 shows the main interface of the application, where the user is provided with options to start recognition or upload a video for testing.

The interface is designed to be user- friendly and clearly presents the system functionality.

Figure 6.3.2 represents the intermediate stage where the uploaded video is being processed by the backend.

The system displays the processing steps such as preprocessing, feature extraction, and sequence inference, indicating that the model pipeline is actively running.

Figure 6.3.3 shows the final output of the system, where the predicted gesture is displayed as text along with a confidence score.

The system is able to correctly recognize gestures such as “had your food” with a high confidence level (99.9%), demonstrating the effectiveness of the trained VideoMAE model.

The inclusion of text output and optional speech output further enhances usability and accessibility for users.

However, certain limitations were observed during testing.

The system performance depends on the quality of input video, lighting conditions, and clarity of gestures.

Additionally, since the model is trained on a limited dataset, it may not generalize well to unseen variations or complex gestures.

Real-time performance was also not consistent, which led to focusing on video-based input for more stable predictions.

Overall, the experimental results validate that the proposed system is capable of recognizing ISL gestures using deep learning techniques.

The system demonstrates the practical feasibility of combining video-based input, transformer models, and web technologies to build an assistive communication tool.

Future improvements can focus on increasing dataset size, improving model generalization, and optimizing inference for real-time applications.

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Figure 6.3.1 : User Interface of the Sign Language Recognition System showing options for video upload and system interaction.

Figure 6.3.2 : Intermediate Processing Stage displaying backend inference steps including preprocessing, feature extraction, and sequence analysis.

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Figure 6.3.3 : Final Prediction Output showing detected ISL gesture as text with confidence score and speech output option.

The developed system successfully demonstrates the integration of deep learning with a web- based interface for Indian Sign Language (ISL) recognition using video input.

By combining a React frontend, FastAPI backend, and a VideoMAE-based model, the system is able to process uploaded videos and generate meaningful gesture predictions in the form of text and optional speech output. The results indicate that sequence-based learning using transformer models is effective in capturing both spatial and temporal features of sign language gestures.

However, the system performance is influenced by factors such as video quality, gesture clarity, and dataset limitations.

While the model performs well on known gesture classes, it may face challenges in handling unseen variations and complex gestures.

Overall, the proposed system proves the feasibility of using deep learning for ISL recognition and provides a strong foundation for future improvements such as better generalization and enhanced real-time performance.

6.3.2 Conclusion

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Performance evaluation plays an important role in analyzing how effectively a deep learning model performs on unseen gesture data.

In this project, the proposed SignAI system is evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

These metrics help in measuring the ability of the model to correctly recognize Indian Sign Language (ISL) gestures and maintain balanced classification performance across multiple gesture classes.

The evaluation is carried out using validation and testing data obtained during the training process.

Since the system is based on a sequence-based learning approach using the VideoMAE transformer model, the evaluation considers both spatial and temporal information present in gesture sequences.

The predictions generated by the model are compared with the actual class labels to determine classification performance and reliability.

Along with numerical evaluation metrics, graphical analysis is also performed to understand the learning behavior of the model.

Training and validation accuracy curves, loss curves, confusion matrix, precision–recall curve, and ROC curve are analyzed to study convergence, classification capability, and generalization performance.

These visualizations help in identifying model behavior such as overfitting, class confusion, and learning stability.

The system is also evaluated from an implementation perspective by testing the complete workflow, including video upload, preprocessing, sequence formation, model inference, and frontend output generation.

The obtained results demonstrate the effectiveness of the proposed transformer-based approach for sentence-level ISL gesture recognition and highlight areas where further improvements can be made in future work.

CHAPTER-7

7.1 INTRODUCTION

RESULT AND CONCLUSION Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 37 2025-26 The comparative analysis was carried out to evaluate the performance of different deep learning approaches used for sign language recognition.

The models considered in this comparison include CNN-based, CNN-LSTM, YOLO-based, and transformer-based VideoMAE architectures.

Each model has its own advantages and limitations depending on how gesture information is processed.

CNN-based models mainly focus on extracting spatial features from individual frames.

These models perform well for static gesture recognition but are less effective in understanding motion continuity across multiple frames.

As a result, their performance decreases when recognizing complex sentence-level gestures.

CNN-LSTM models improve upon CNN architectures by combining spatial feature extraction with sequential learning. The LSTM component helps in learning temporal dependencies between frames. However, these models still face challenges in capturing long-range relationships and subtle gesture transitions present in continuous sign language sequences. YOLO-based models achieve higher accuracy because they are highly effective in detecting hands and gesture-related regions within frames. These models are mainly designed for object detection tasks and perform well in frame-level gesture identification. However, they focus more on localization and less on understanding the complete temporal flow of gestures. In contrast, the proposed VideoMAE model follows a sequence-based transformer architecture that processes entire video sequences simultaneously. Using self-attention mechanisms, the model learns both spatial and temporal relationships across frames more effectively. Although the overall accuracy of VideoMAE is lower than YOLO- based detection models, it provides better understanding of gesture continuity and sentence-level motion patterns. The higher AUC-ROC value achieved by VideoMAE indicates strong class separability and stable classification performance across multiple gesture classes.

7.2 COMPARATIVE ANALYSIS

Model Type Accuracy (%) Precision (%) Recall (%) F1-Score (%) AUC-ROC CNN- Based Spatial 65 67 64 65 0.83 CNN + LSTM Hybrid 69 70 68 69 0.85 YOLO- Based Detection 85 87.5 85 85 0.88 VideoMAE Transformer 74 77 74 74 0.98 Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 38 2025-26 The transformer-based architecture also improves generalization capability and robustness under different gesture conditions. Therefore, VideoMAE is more suitable for real-world sentence-level Indian Sign Language recognition compared to traditional frame-based approaches.

7.3 RESULT

The proposed SignAI system was successfully developed and implemented for sentence-level Indian Sign Language (ISL) recognition using a transformer-based VideoMAE architecture. The system integrates frontend video upload functionality, backend preprocessing, deep learning-based inference, and text output generation into a complete end-to-end framework. The developed preprocessing pipeline effectively converts raw video input into structured frame sequences suitable for sequence-based learning. Video frames are extracted, resized to 224×224 224 \times 224 224\times 224, normalized, and arranged into fixed-length sequences of 16 frames. This preprocessing stage helps preserve temporal information required for gesture understanding. The trained VideoMAE model demonstrates effective learning of both spatial and temporal gesture features from video sequences. Experimental evaluation performed on the ISL- CSLTR dataset shows that the model achieves a validation accuracy of approximately 74%, with precision, recall, and F1-score values of 77%, 74%, and 74% respectively. The obtained AUC-ROC score of 0.98 indicates strong classification capability and reliable class separation.

Table 7.2.1: Performance Comparison of Different Approaches

Training and validation performance curves show stable convergence during training, while the confusion matrix demonstrates that most gesture classes are correctly classified.

The Precision–Recall and ROC curves further confirm the robustness of the proposed model for multi-class gesture recognition tasks.

The frontend system built using React successfully allows users to upload gesture videos and view prediction outputs through an interactive interface.

The backend developed using FastAPI performs preprocessing and model inference efficiently.

The predicted gesture output is displayed as text along with confidence scores, and optional speech output is also supported for improved accessibility.

Overall, the project results demonstrate that transformer-based sequence learning approaches such as VideoMAE are effective for sentence-level ISL recognition.

The system provides a reliable framework for gesture understanding and can serve as a foundation for future real-world assistive communication applications.

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Figure 7.3.1 : Performance metrics of the proposed ISL recognition system showing accuracy, precision, recall, and F1-score.

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Figure 7.3.2 : Performance metrics of the proposed ISL recognition system showing accuracy, precision, recall, and F1-score.

This project presents the design and implementation of an Indian Sign Language recognition system using deep learning and web technologies.

The system successfully integrates a VideoMAE-based model with a React frontend and FastAPI backend to create an end-to-end pipeline for gesture recognition from video input.

The use of sequence-based learning enables the model to capture both spatial and temporal information, resulting in effective gesture classification.

The experimental results demonstrate that the system achieves satisfactory performance with balanced accuracy, precision, recall, and F1-score.

The system is capable of recognizing a wide range of gestures and generating meaningful outputs in the form of text and optional speech.

However, certain limitations such as sensitivity to input quality, dataset constraints, and generalization challenges were observed during testing.

Overall, the project validates the feasibility of using transformer-based models for sign language recognition and highlights the importance of temporal modeling in gesture understanding.

The developed system serves as a strong foundation for future enhancements, including improved accuracy, larger datasets, and real-time implementation, making it a valuable step toward accessible and intelligent communication systems.

7.4 CONCLUSION

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 41 2025-26 The current system can be further improved by enhancing the dataset with more diverse gesture samples, including variations in lighting, background, and user styles to improve model generalization.

Future work can also focus on enabling real-time gesture recognition using live webcam input and optimizing the model for faster inference.

Additionally, integrating more advanced transformer architectures or multimodal approaches (such as combining gesture with facial expressions) can improve accuracy for complex gestures.

Expanding the system to support multiple sign languages and improving robustness for similar gesture patterns

will make the application more practical for real-world communication.

7.5 Future Work

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CHAPTER-9

9.1 UNIT TESTING

Software Testing and Result Unit testing ensures that each individual component of the SignAI system works correctly when tested independently.

This helps in identifying errors in preprocessing, model inference, and data handling at an early stage before full system integration.

Scope in This Project: Testing video frame extraction logic for correct frame generation.

Verifying resizing and normalization of frames to 224×224 format.

Testing sequence formation logic (16 frames per sequence).

Validating VideoMAE model inference on sample inputs.

Checking classification output for correct gesture labels.

Tools Used: Python scripts for preprocessing validation.

PyTorch for model inference testing.

OpenCV for frame extraction and image processing validation.

Outcome: All individual modules produced correct outputs during isolated testing.

This ensured proper functionality of preprocessing and model components and reduced errors during integration.

9.2 INTEGRATION TESTING

Integration testing focuses on validating the interaction between different modules of the SignAI system, including frontend, backend, and model inference components.

Scope in This Project: Testing complete flow: video upload → preprocessing → model inference → output display.

Verifying communication between React frontend and FastAPI backend.

Ensuring API correctly handles input video and returns predictions.

Validating synchronization between preprocessing and model pipeline.

Tools Used: FastAPI for backend API testing.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 38 2025-26 React.js for frontend integration testing.

Python test scripts for API validation.

Outcome: All integrated modules communicated effectively and performed as expected.

Issues related to data flow and API responses were identified and resolved, ensuring smooth system operation.

9.3 SYSTEM TESTING

System testing validates the overall performance of the complete SignAI system under real-world usage conditions.

Scope in This Project: End-to-end testing from video input to final output generation.

Validation of gesture recognition accuracy across different inputs.

Testing system behavior under multiple user interactions.

Evaluating system response under different lighting and gesture variations.

Tools Used: Web browser for full system execution.

React interface for user interaction testing.

FastAPI backend for processing requests.

Testing Criteria: Functional Testing: Verified correct functioning of all modules such as preprocessing, model inference, and frontend display.

Accuracy Testing: Ensured that predicted gesture labels match expected outputs for test samples.

Performance Testing: Evaluated system efficiency and response time during video processing and prediction.

Usability Testing: Checked ease of use of the web interface for uploading videos and viewing results.

Robustness Testing: Tested system performance under variations such as lighting conditions, background noise, and gesture differences.

Outcome: The system successfully performed end-to-end gesture recognition.

The pipeline from input to output worked reliably, and the system handled different test cases effectively.

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Figure : Planned project development timeline

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 43 2025-26 The proposed Sign Language Recognition System (SignAI) aligns with the United Nations Sustainable Development Goals (SDGs) by addressing communication accessibility, inclusion, and technological innovation.

The relevant SDGs and their mapping to this project are given below: SDG 3: Good Health and Well-Being Ensure healthy lives and promote well-being for all at all ages.

This project supports SDG 3 by providing an assistive communication solution for individuals with hearing and speech impairments.

By enabling automatic recognition of sign language gestures and converting them into text and speech, the system contributes to improving mental well-being, independence, and quality of life.

SDG 4: Quality Education Ensure inclusive and equitable quality education and promote lifelong learning opportunities.

The proposed system can be used as an assistive tool for students with hearing or speech disabilities, helping them actively participate in classroom environments.

It also promotes research and innovation in interdisciplinary fields such as artificial intelligence, computer vision, and human-computer interaction.

SDG 9: Industry, Innovation and Infrastructure Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

By integrating deep learning models such as VideoMAE with modern web technologies, the project encourages technological innovation and the development of intelligent systems that can be extended to real-world communication and accessibility applications.

SDG 10: Reduced Inequalities Reduce inequality within and among countries.

The development of a cost-effective and accessible sign language recognition system aims to bridge the communication gap between deaf and hearing communities.

This helps in reducing inequality in access to communication tools and digital services.

SDG 17: Partnerships for the Goals (Indirect Contribution) Strengthen the means of implementation and revitalize global partnerships.

The project leverages publicly available datasets, open-source frameworks, and collaborative development practices, promoting knowledge sharing and partnerships across academic and technological communities.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 44 2025-26 CO-PO MAPPING Program Outcomes (POs) PO1 – Engineering Knowledge Applies knowledge of artificial intelligence, computer vision, deep learning, and software engineering to process video data and develop the sign language recognition system.

PO2 – Problem Analysis Analyzes challenges in recognizing Indian Sign Language gestures from video sequences.

Identifies suitable sequence-based learning approaches to handle gesture variations and temporal dependencies.
PO3 – Design/Development of Solutions Designs an integrated sign language recognition system combining preprocessing, VideoMAE-based deep learning, backend APIs, and frontend interfaces for effective gesture recognition.

PO4 – Conduct Investigations of Complex Problems Conducts experiments on ISL gesture datasets and evaluates multiple approaches such as CNN, CNN-LSTM, YOLO, and VideoMAE using different performance metrics.

PO5 – Modern Tool Usage Uses modern tools and technologies such as Python, PyTorch, OpenCV, FastAPI, React, and transformer-based deep learning frameworks for system development.

PO6 – The Engineer and Society Addresses communication difficulties faced by the deaf and mute community through the development of an assistive AI-based communication system.

PO7 – Environment and Sustainability Promotes the development of digital assistive technologies that improve accessibility and reduce dependency on manual communication support systems.

PO8 – Ethics Ensures responsible handling of user data and ethical use of artificial intelligence models while maintaining fairness in gesture recognition.

PO9 – Individual and Team Work Works effectively both individually and collaboratively during dataset preparation, preprocessing, model training, integration, and testing phases.

PO10 – Communication Communicates technical concepts, project methodology, results, and system outputs effectively through reports, presentations, and demonstrations.

PO11 – Project Management and Finance Plans and manages project activities including data collection, preprocessing, model development, deployment, and testing within the project timeline.

PO12 – Life-long Learning Learns and adapts to emerging technologies such as transformers, VideoMAE, deep learning, and AI-based vision systems for continuous improvement.

Course Outcomes (COs) CO1 Analyze the problem of Indian Sign Language recognition by studying existing research, identifying limitations in traditional approaches, and defining suitable methodologies using transformer-based deep learning techniques for assistive communication systems.

CO2 Design, develop, and implement a complete sign language recognition system using VideoMAE, sequence-based preprocessing, backend APIs, and frontend interfaces to generate accurate text output from gesture videos.

CO3 Evaluate the performance of the proposed SignAI system using accuracy, precision, recall, F1-score, confusion matrix, PR curve, and ROC analysis, and present the obtained results through reports, visualizations, and system demonstrations.

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SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 45 2025-26 CO \ PO/
PSO PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO1 0 PO1 1 PSO 1 PSO 2 CO1 3 3 3 2 2 1 2 2 2 1 1 2 1 CO2
3 2 3 1 2 1 1 2 2 2 1 2 1 CO3 3 2 2 1 2 2 3 2 2 1 1 1 1 Dept of IS&E, PESCE, Mandya.

SignAI: Indian Sign Language Recognition Using VideoMAE-Based Vision Transformers 46 2025-26 CO–PO &
POS Mapping Table Program Specific Outcomes (PSOs) PSO1 – Software Development and Programming
Skills Applies programming and software development skills to build preprocessing pipelines, AI models,
backend APIs, and frontend interfaces for the proposed system.

PSO2 – Application of Emerging Technologies Applies transformer-based deep learning, computer vision, and
sequence modeling techniques to solve real-world sign language communication problems.

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